



SEMESTER 1 EXAMINATIONS 2023/2024

MODULE: CA4023 - Natural Language Technologies

PROGRAMME(S):

DS	BSc in Data Science
MCM	M.Sc. in Computing

YEAR OF STUDY: 1,4

EXAMINER(S):

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TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer Question 5 and three other questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1

[TOTAL MARKS: 25]

Q 1(a)

[9 Marks]

Consider the following training corpus containing restaurant reviews. The sentiment label of each review is in brackets beside it.

Avoid this place (negative)

Avoid the chicken wings (negative)

This place is awesome (positive)

What a lovely atmosphere (positive)

Show, step by step, how a Naïve Bayes sentiment classifier which uses words as features and which is trained on the above corpus would classify the following restaurant review as either positive or negative:

This place serves chicken wings

Q 1(b)

[5 Marks]

Using Q1(a) as an example, explain why a Naïve Bayes classifier can sometimes misclassify the sentiment of reviews.

Q 1(c)

[5 Marks]

Naïve Bayes and Logistic Regression are commonly used for document classification. One is a **discriminative** classifier and one is a **generative** classifier. State which is discriminative and which is generative, and explain the difference between the two types of classifier.

Q 1(d)

[6 Marks]

Explain how a recurrent neural network (RNN) can be used to classify a document. Sketch the architecture of an RNN which classifies documents.

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[8 Marks]

Given the following training corpus for part-of-speech tagging, state the **emission probabilities** that would be learned from this corpus by a Hidden Markov Model (HMM) part-of-speech tagger:

*Dog/NOUN bites/VERB man/NOUN
Man/NOUN feeds/VERB dog/NOUN*

Q 2(b)

[3 Marks]

Apart from part-of-speech tagging, name one other sequence labelling task and say why it counts as sequence labelling.

Q 2(c)

[6 Marks]

Name three pieces of information that can be used by a **linear-chain Conditional Random Field (CRF)** sequence labeller that cannot be used by a HMM sequence labeller. For each piece of information, say why it might be useful for a particular sequence labelling task.

Q 2(d)

[8 Marks]

What is the difference between **document classification** and **sequence labelling**? Outline the difference between how the two types of tasks are carried out using a recurrent neural network (RNN)?

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[8 Marks]

Consider the following training corpus:

<s> x y z </s>
<s> y </s>
<s> y </s>
<s> x z </s>
<s> y x x </s>

What is the probability of the following four strings according to an unsmoothed a) **unigram** language model, and b) **bigram** language model, trained on the above corpus?

1. <s> x </s>
2. <s> y </s>
3. <s> z </s>
4. <s> y x y </s>

Q 3(b)

[4 Marks]

Explain, using an example, why **smoothing** is needed for n-gram language modelling and briefly explain one smoothing technique.

Q 3(c)

[9 Marks]

Explain how the following three types of neural language model take context into account when predicting the next word in a sequence of words:

1. A **feedforward** neural language model
2. A **recurrent** neural language model
3. A **transformer** neural language model

Q 3(d)

[4 Marks]

State one difference and one similarity between the **BERT** and **GPT** language models.

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 25]

Q 4(a)

[6 Marks]

Explain, using examples, two ways in which word vectors are useful for Natural Language Processing.

Q 4(b)

[7 Marks]

Consider the following word co-occurrence matrix and then answer the following questions:

	storm	debris	workshop	teaching
winds	50	28	1	3
electricity	45	14	5	6
trees	27	36	2	3
curriculum	1	0	17	35

1. What does the 28 in the (*winds*,*debris*) cell refer to?
2. What is the vector for the word *trees*?
3. How would you calculate the similarity between the words *curriculum* and *trees*?
4. How would you calculate the Positive Pointwise Mutual Information between the words *electricity* and *workshop*?

Q 4(c)

[6 Marks]

Explain the advantage of

- 1) **dense** word vectors over **sparse** word vectors.
- 2) **dynamic** (or contextualised) dense vectors over **static** (or non-contextualised) dense vectors.

Q 4(d)

[6 Marks]

Name two hyper-parameters that are used in the training of **Word2Vec** (skipgram negative sampling) and briefly describe the role of each.

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Q 5(a)

[16 Marks]

Under each of the following categories, describe and explain what you consider the most pressing ethical concern around Large Language Models:

1. Environment
2. Economy
3. Education
4. The Arts

Q 5(b)

[9 Marks]

For each concern mentioned in Q 5(a), outline what you think should be done to address the concern.

[End of Question 5]

[END OF EXAM]